

Alphabet

NASDAQ : \$4366.3B mkt cap · 12.09B sh · \$126.8B cash, \$48.5B net debt · \$126.8B cash vs \$48.5B debt (~\$78.3B net cash); ~12.09B shares (Class A+B+C), buyback + a small dividend; Revenue \$402.8B FY25 (+15%); GAAP op margin 32.0%; FCF \$73.3B; capex ~\$91B and guided to ~\$190B in FY26; Up ~28% YTD and ~130-160% TTM to ~\$360 / ~\$4.35T cap; ATH close \$402.38 (May-2026), re-rated on owning the full AI stack

DCF

\$361.21

THE CENTRAL QUESTION

Does Alphabet's full AI stack (TPUs→Gemini→Cloud→apps) defend and monetize Search faster than generative-AI assistants erode its query and ad economics?

Alphabet trades at ~\$4.35T (~\$360, +28% YTD and ~130-160% TTM) - roughly 60x EV/FCF on \$402.8B revenue growing ~15%, 32% operating margins, net cash, \$73.3B FCF before a capex ramp to ~\$190B. Cloud just re-accelerated to +63% with a ~\$462B backlog and the AI stack re-rated the name, but Search is still ~60%+ of revenue and the price embeds the win. The DCF asks whether Search holds its economics through the generative-AI shift while Cloud compounds. The finding lands -21.4% (weighted ~\$286): the modal base sits right at spot, so the run priced in the franchise and then some.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$282.54 expected fair value today
-21.8% vs spot \$361.21

FORWARD COMPOUNDED VALUE

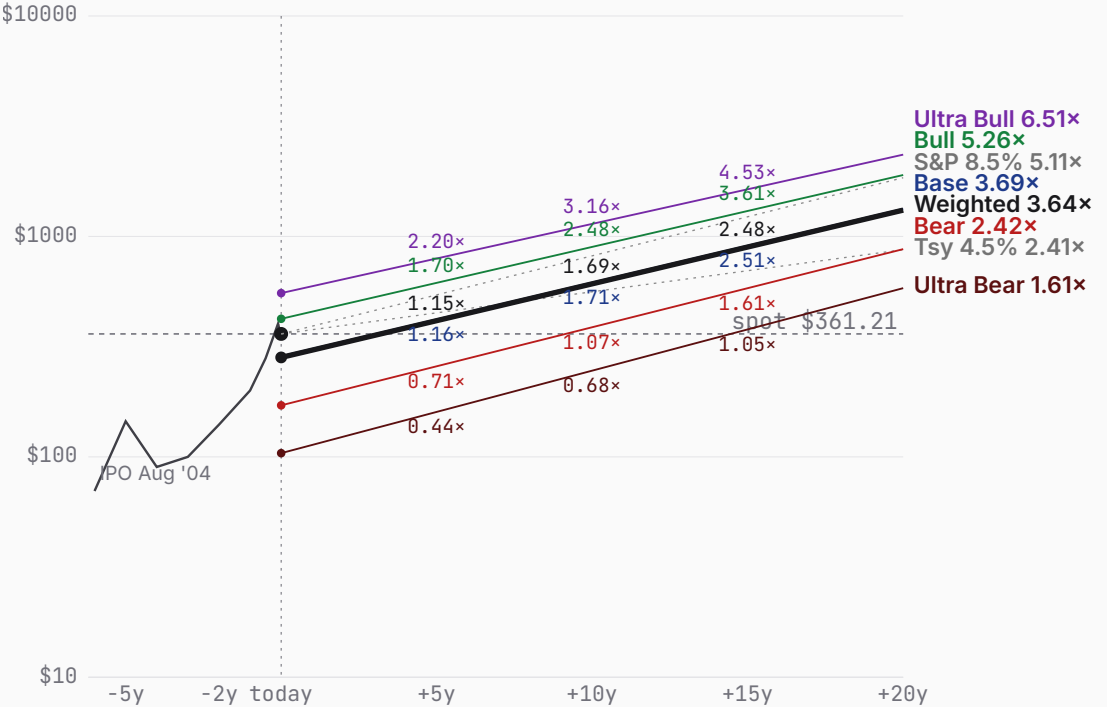
+5y	+10y	+15y	+20y
\$414.92	\$609.49	\$895.58	\$1,316.37
1.15× spot	1.69× spot	2.48× spot	3.64× spot

WEIGHTING RATIONALE

- Ultra Bear 13% AI assistants erode Search economics; ~\$104 (-71%).
- Bear 25% Search slows; capex caps returns; ~\$171 (-52%).
- Base 34% Search defends; Cloud compounds; ~\$286 (-20%).
- Bull 20% AI stack compounds; Cloud re-rates; ~\$423 (+18%).
- Ultra Bull 8% Full-stack AI platform re-rates; ~\$553 (+54%).

Bull (20%) + Ultra Bull (8%) together contribute \$128.85 — 46% of the \$282.54 expected value despite 28% combined probability. Today's spot (\$361.21) sits 28% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$103.95	BEAR	\$171.32	BASE	\$286.32	BULL	\$422.99	ULTRA BULL	\$553.18
	-71.2% vs spot		-52.6% vs spot		-20.7% vs spot		+17.1% vs spot		+53.1% vs spot
CAGR +4.8% WACC 9.0% CAGR +4.8% Prob 13%		CAGR +9.6% WACC 8.5% CAGR +9.6% Prob 25%		CAGR +13.4% WACC 8.0% CAGR +13.4% Prob 34%		CAGR +16.2% WACC 7.8% CAGR +16.2% Prob 20%		CAGR +18.7% WACC 7.5% CAGR +18.7% Prob 8%	
AI assistants gut Search economics.		Search slows; capex caps returns.		Search defends; Cloud compounds.		AI stack compounds; Cloud re-rates.		Full-stack AI platform; multiple re-rates.	



Alphabet scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 13% Bear 25% Base 34% Bull 20% Ultra Bull 8% · Weighted expected \$282.54 (-21.8% vs spot)

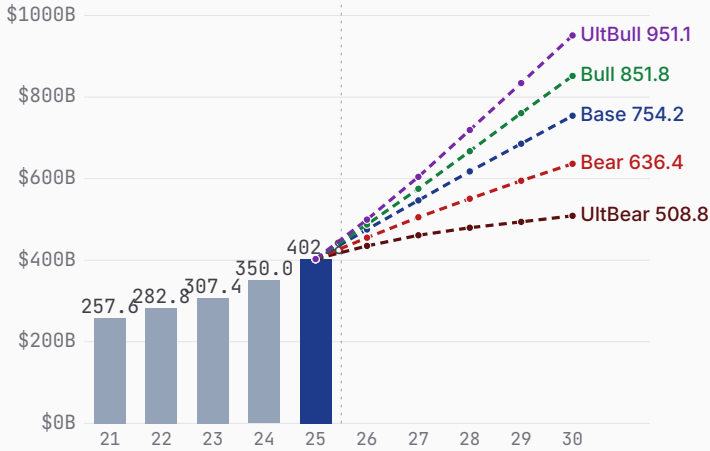
ULTRA BEAR	\$103.95	BEAR	\$171.32	BASE	\$286.32	BULL	\$422.99	ULTRA BULL	\$553.18
	-71.2% vs spot		-52.6% vs spot		-20.7% vs spot		+17.1% vs spot		+53.1% vs spot
Probability 13%		Probability 25%		Probability 34%		Probability 20%		Probability 8%	
AI assistants gut Search economics.		Search slows; capex caps returns.		Search defends; Cloud compounds.		AI stack compounds; Cloud re-rates.		Full-stack AI platform; multiple re-rates.	
WHY 13%		WHY 25%		WHY 34%		WHY 20%		WHY 8%	
The genuine bear: Search is ~60%+ of revenue, the capex commitment is now ~\$190B and 'significantly higher' in 2027, and chatbots are a live substitution threat. 13% weight on the AI shift impairing the core.		Search holds share but the shift compresses ad economics while ~\$190B capex weighs on returns - the plausible 'defends but de-rates' path. 25% given the antitrust overhang and the capex trajectory.		Requires Search to monetize AI at near-parity and Cloud to keep compounding while capex stays productive - consistent with Q1'26 (Search +19%, Cloud +63%, margin doubling). 34% as the central, near-spot outcome.		Cloud's +63% and doubling margin plus a defended, AI-monetized Search make a genuine compounding case credible. 20% weight on the full AI stack delivering on both legs.		Everything works - Search monetizes AI above parity AND Cloud compounds AND TPUs/Gemini/Waymo add value AND the multiple holds. ~8%, the long-duration tail.	
WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS	
Generative-AI assistants take real query share and lower ad load: Search growth stalls toward low-single-digit, AI Overviews monetize below legacy CPCs, and the ~\$190B capex ramp lands while revenue decelerates - so FCF margin sinks to ~15-17% off the ~18% base. Cloud keeps growing but can't offset a structurally impaired core.		Search defends volume but ad load and CPCs drift lower as queries route through AI surfaces, growth decelerates to high-single-digit, and the capex ramp holds FCF margin near 18-20% even as Cloud scales. The AI stack works but the blended economics are lower than the past decade's.		The modal path: Search defends and monetizes the AI transition at roughly parity (Q1'26's +19% argues defense is winning), Cloud keeps compounding off the ~\$462B backlog with expanding margins, and FCF margin recovers from the capex trough toward ~24% as the ~\$190B spend builds capacity that fills. Revenue compounds low-double-digit to ~\$680B.		The full stack pays off: TPUs lower the cost of serving Gemini, Cloud sustains hyper-growth and operating leverage off the backlog, AI Mode/Overviews lift Search monetization above legacy rates, and FCF margin climbs toward ~27% once the capex wave is absorbed. Revenue compounds mid-teens to ~\$730B.		Everything compounds: Gemini + AI Mode expand Search economics, Cloud becomes a durable second franchise at scale, Waymo and the mode/TPU stack add optionality, and FCF margin reaches ~29% as the capex cycle turns into high-return capacity. Revenue compounds high-teens to ~\$790B.	
A franchise whose primary cash engine is being disrupted earns a low-growth, ~16x terminal multiple → DCF ~\$104 (-71%). The net cash sets a floor but can't offset a Search that loses its economics while capex stays committed.		DCF ~\$171 (-52%) - a still-dominant platform that grows slower and converts less of revenue to free cash flow at ~19x FCF is worth well below today's price; the post-run multiple already prices the win.		DCF ~\$286 (-20%) - the modal case lands right at spot. A franchise growing low-double-digit at a recovering ~24% FCF margin and a ~23x multiple is worth about today's price; the 2026 run priced the franchise in, leaving little margin of safety at the modal outcome.		DCF ~\$423 (+18%) - if Search not only defends but Cloud + the AI stack compound into a second profit engine, the franchise is worth meaningfully above spot at a ~27x multiple; this is the bull the re-rating is paying for.		DCF ~\$553 (+54%) - Alphabet re-rates as the broadest full-stack AI platform; ~8% probability, the asymmetric upside that owning chips through apps could deliver.	

Alphabet business snapshot

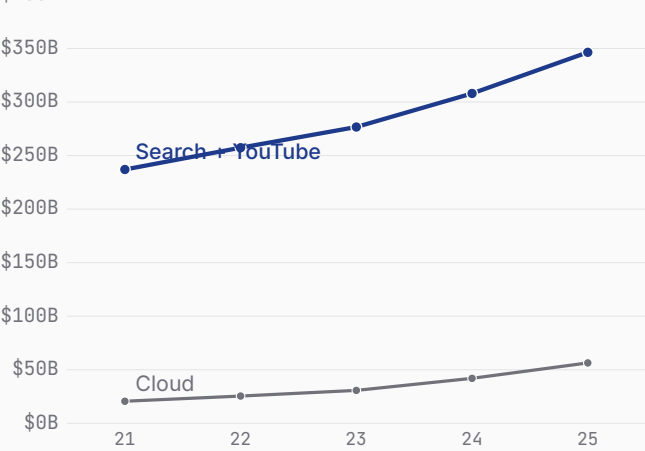
Bear \$171.32 Base \$286.32 Bull \$422.99

FY21-FY25 history + FY26-FY30 scenario projections · fiscal years end Dec · Q1 2026 results

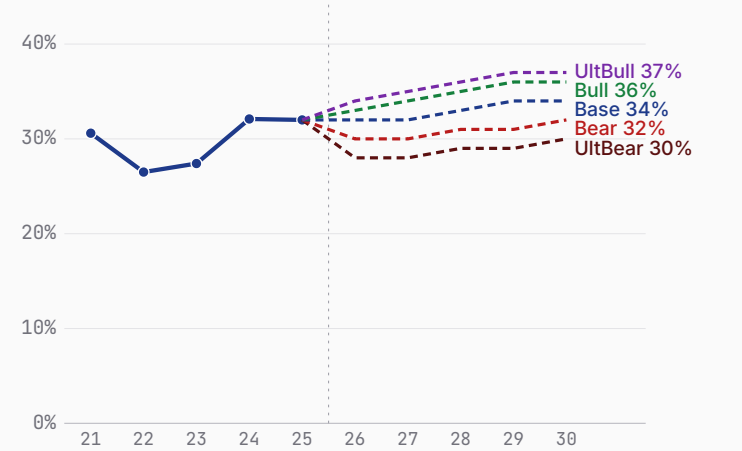
Revenue (\$B) — history + scenarios to FY30



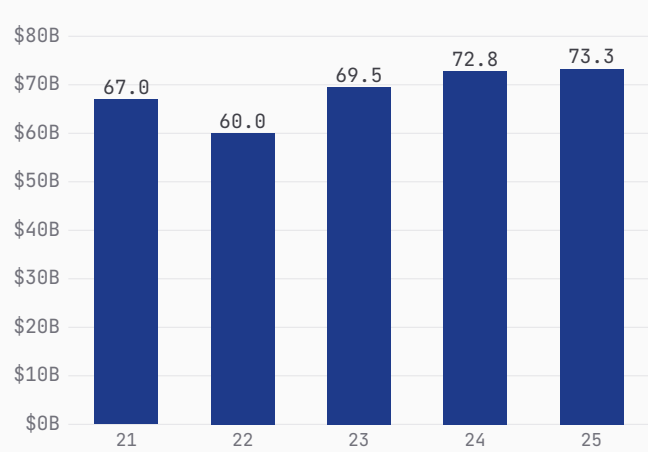
Revenue by segment (\$B)



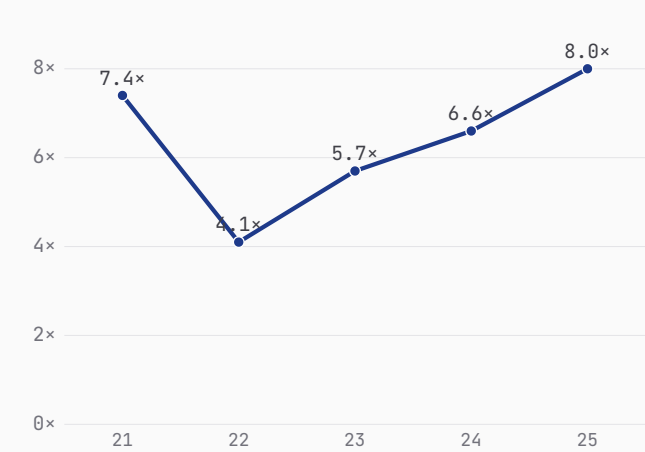
Op margin — history + scenarios



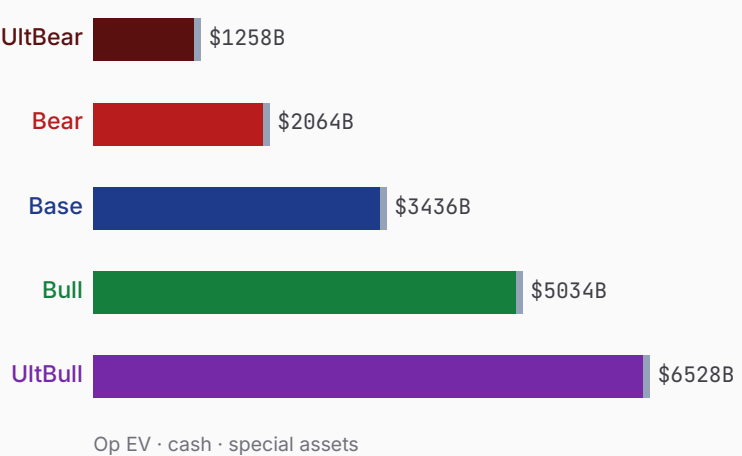
Free cash flow (\$B) — history



EV / Revenue — historical



Equity build (Op EV + net cash)



Sources: SEC XBRL company facts (CIK 1652044; revenue, GAAP operating income/margin, cash flow, debt), Alphabet Q1 2026 release (revenue \$109.9B/+22%, op margin 36.1%, Search & other +19% to \$60.4B, Cloud +63% to \$20.0B at 32.9% margin, backlog ~\$462B, FY26 capex guided ~\$190B). Revenue modeled as a growth-rate path off FY25; FCF = revenue x FCF margin; Gordon terminal at scenario WACC. EV/sales & EV/FCF vs mega-cap platform peers (MSFT/META/AMZN).

Alphabet 7 Powers · the moat, scored

Bear \$171.32 Base \$286.32 Bull \$422.99

Arena. Search + digital advertising and cloud/AI infrastructure - Alphabet (Search, YouTube, Cloud, the TPU→Gemini→apps stack) vs Microsoft/OpenAI, Amazon, Meta, and chatbot challengers; auditing entrenched network/scale/cornered-resource Powers against a generative-AI substitution and a live antitrust threat.

POWER AUDIT
dominant-power durability: high

POWER AUDIT · 7 POWERS

Scale Economies 	Massive fixed-cost leverage across the index, ad systems, YouTube, and global data centers; 32% operating margin at \$403B revenue - few can match the spend per query.
Network Economies · thesis leans here 	Search's two-sided advertiser/user loop, YouTube's creator/viewer flywheel, and Android/Play ecosystems - more users sharpen results and pricing, a genuine network Power.
Counter-Positioning 	The incumbent now being counter-positioned by AI-native assistants (ChatGPT), not the disruptor; AI Overviews is a defensive retrofit, not counter-positioning.
Switching Costs 	Workspace/Cloud contracts and the ~\$462B backlog lock enterprises in; consumer Search switching is low, but Android/Chrome/account defaults raise friction.
Branding 	'Google' is a verb and a default trust signal, but brand alone won't hold queries if an assistant answers better - durable yet not the binding Power.
Cornered Resource 	The crawl index, decades of query/click data, custom TPUs, and frontier Gemini models - assets rivals cannot quickly replicate; the data + silicon edge is the deepest moat.
Process Power 	Deep ML/infra engineering capability, but increasingly a contested field rather than a unique repeatable process.

PEER SET

Microsoft / OpenAI Azure + Copilot + OpenAI partnership attack both Cloud and Search via AI assistants; the primary full-stack rival.	16% gr · 45% mgn · ~35x P/E
Amazon (AWS) Cloud market leader; the EV/FCF and Cloud-growth benchmark Alphabet's Cloud is measured against.	15% gr · 11% mgn · ~30x EV/FCF
Meta Platforms Ad-budget rival and AI-compute peer; competes for advertiser dollars and AI talent/capex.	20% gr · 41% mgn · ~24x EV/FCF
OpenAI / Anthropic Frontier-model labs whose assistants are the direct generative-AI substitution threat to Search query volume.	· private

THREAT VECTORS · FALSIFIERS

Network Economies · ChatGPT / AI-native assistants falsifier: Search & other growth decelerates toward low-single-digit or query volume visibly leaks to chatbots, breaking the user/advertiser loop.
Cornered Resource · Frontier labs + merchant silicon (NVIDIA) falsifier: Rivals match Gemini quality and serving cost, eroding the data/TPU advantage as a differentiator.
Scale Economies · DOJ / ad-tech antitrust remedy falsifier: An adverse ad-tech ruling forces an AdX/DFP divestiture or default-distribution bans dent Search's scale and reach.

COMPETITIVE READ

Alphabet's network and scale Powers are deeply entrenched - a two-sided Search/advertiser loop, YouTube's flywheel, and a cornered-resource stack (index, query data, custom TPUs, Gemini) few rivals can replicate. The Audit: network_economies is durable (high); the live threats are generative-AI assistants substituting for ~60%+-of-revenue Search and an antitrust overhang (DOJ search appeal, a pending ad-tech divestiture) the DCF doesn't price. Q1'26 (Search +19%, Cloud +63%) says the defense is winning - but the ~28% YTD run re-rated the name to ~60x EV/FCF, so the modal case lands at spot: -21.4%, a durable franchise priced for the win.

Alphabet show your work

Bear \$171.32 Base \$286.32 Bull \$422.99 · Weighted \$282.54 (-21.8%)

ULTRA BEAR					BEAR					BASE					BULL					ULTRA BULL				
\$103.95					\$171.32					\$286.32					\$422.99					\$553.18				
ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS				
5y revenue CAGR (%)					+4.8					+9.6					+13.4					+16.2				
Year-5 op margin (%)					30.0					32.0					34.0					36.0				
WACC (%)					9.0					8.5					8.0					7.8				
Terminal growth (%)					2.5					3.0					3.5					4.0				
5y revenue CAGR (%)					+4.8					+9.6					+13.4					+16.2				
Starting revenue (\$B)					402.80					402.80					402.80					402.80				
Scenario probability (%)					13					25					34					20				
5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B				
FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF
FY26	435.02	+28%	+65.25	+59.87	FY26	455.16	+30%	+77.38	+71.32	FY26	475.30	+32%	+90.31	+83.62	FY26	487.39	+33%	+97.48	+90.42	FY26	499.47	+34%	+104.89	+97.57
FY27	461.13	+28%	+69.17	+58.22	FY27	505.23	+30%	+90.94	+77.25	FY27	546.60	+32%	+109.32	+93.72	FY27	575.12	+34%	+126.53	+108.88	FY27	604.36	+35%	+145.05	+125.51
FY28	479.57	+29%	+76.73	+59.25	FY28	550.70	+31%	+104.63	+81.92	FY28	617.66	+33%	+135.88	+107.87	FY28	667.14	+35%	+160.11	+127.81	FY28	719.19	+36%	+186.99	+150.52
FY29	493.96	+29%	+83.97	+59.49	FY29	594.76	+31%	+118.95	+85.83	FY29	685.60	+34%	+157.69	+115.91	FY29	760.54	+36%	+197.74	+146.43	FY29	834.26	+37%	+233.59	+174.91
FY30	508.78	+30%	+86.49	+56.21	FY30	636.39	+32%	+127.28	+84.65	FY30	754.16	+34%	+181.00	+123.18	FY30	851.80	+36%	+229.99	+157.98	FY30	951.06	+37%	+275.81	+192.12
Σ PV explicit FCF					+293.04					+400.96					+631.52					+740.63				
+ PV terminal (g 2.5%)					886.45					1585.18					4323.72					5708.58				
EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE				
Operating EV					\$1179.49B					\$1986.14B					\$4955.24B					\$6449.21B				
+ Cash & investments					\$126.80B					\$126.80B					\$126.80B					\$126.80B				
– Net debt					\$48.50B					\$48.50B					\$48.50B					\$48.50B				
= Equity value					\$1257.79B					\$2064.44B					\$5033.54B					\$6527.51B				
FY30 shares (M)					12,100					12,050					11,900					11,800				
= Expected per share					\$103.95					\$171.32					\$422.99					\$553.18				
FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT				
+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y	
\$159.94	\$246.09	\$378.64	\$582.58		\$257.61	\$387.35	\$582.44	\$875.80		\$420.70	\$618.14	\$908.26	\$1,334.53		\$615.78	\$896.43	\$1,305.00	\$1,899.79		\$794.16	\$1,140.12	\$1,636.79	\$2,349.83	
0.44×	0.68×	1.05×	1.61×		0.71×	1.07×	1.61×	2.42×		1.16×	1.71×	2.51×	3.69×		1.70×	2.48×	3.61×	5.26×		2.20×	3.16×	4.53×	6.51×	



Alphabet People · Opportunity · Context · Deal

Bear \$171.32 Base \$286.32 Bull \$422.99

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Sundar Pichai

6 yrs in seat

6%
insider ownership

MEDIUM
key-person risk

Capital allocation (realized)
Capital return ramping off a long no-payout history: initiated the first-ever dividend April 2024 (\$0.20/qtr, raised 5% to \$0.21 in 2025), alongside two consecutive \$70B buyback authorizations (April 2024 + April 2025). Simultaneously a record capex build — FY2025 guidance raised \$75B to \$85B to \$91-93B for AI infrastructure (TPUs/GPUs, data centers). Selective tuck-in M&A; SBC a multi-billion annual cost.

Incentive alignment
Non-founder CEO paid mostly in equity: FY2024 total comp \$10.73M on a \$2M base, the remainder equity. PSUs granted on a ~3-year cycle (2024 was an off-grant year; the 2022 grant pushed total comp to ~\$226M); long-term grants carry performance/TSR conditions. Pichai holds Class A/C from comp, not the super-voting Class B.

GOVERNANCE FLAGS

triple-class: Class A (1 vote, GOOGL) · Class B (10 votes, founder-held, unlisted) · Class C (0 votes, GOOG)

founders Page + Brin control ~52.7% of votes via Class B (Page 27.1% + Brin 25.2%) on ~6% economics — majority voting control

CEO Pichai operates but holds no Class B; founders, not the CEO, hold control and cannot be outvoted by outside holders

public Class C (GOOG) carries no vote at all

PEOPLE READ

Founders Page and Brin retain ~52.7% combined voting control via 10-vote Class B on roughly 6% economic ownership, while non-founder CEO Pichai (Alphabet CEO since Dec 2019) runs operations; capital return is ramping (first dividend 2024, repeated \$70B buybacks) against a record AI capex build.

THE FOUR LEGS

People
observable composite · 1–5

Opportunity
The scenario distribution · the Page-3 business snapshot · the Arthur Indicator

Context
The 7 Powers analysis (Power Audit)

Deal
Open-market purchase = the deal: entry price vs. the scenario distribution (the ~21.4% finding) + position sizing.



Alphabet supporting analysis · disclaimers · glossary

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PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1 **The run priced the franchise in.**
After +28% YTD / ~130-160% TTM the name is ~60x EV/FCF and the modal DCF is ~-20% - the AI stack is real, but the price already embeds the win, leaving thin margin of safety.
- 2 **Cloud at +63% with doubling margins is the new engine.**
Cloud hit \$20.0B/+63% at 32.9% operating margin (from 17.8%) with a ~\$462B backlog ~2x QoQ - a credible second franchise that the bull/ultra-bull cases lean on.
- 3 **Search defense is winning so far.**
Q1'26 Search & other still grew +19% to \$60.4B with AI Overviews at 2.5B MAU monetizing near parity - the central bear (query/CPC leakage) is not yet visible in the numbers.
- 4 **The ~\$190B capex is the FCF swing.**
Capex guided to ~\$190B FY26 and higher in 2027 crushes near-term FCF; the bull needs it to convert to high-return capacity.
- 5 **Antitrust is the unpriced tail.**
The DOJ search remedy (default bans + data-sharing, on appeal) and a pending ad-tech remedy (possible AdX/DFP divestiture) sit outside the DCF.

FALSIFICATION TRIGGERS

Bull validation Cloud growth holds >40% with margin above 30% · Search & other stays >+15% as AI Overviews/AI Mode monetize · FCF margin recovers above ~22% as capex is absorbed · backlog keeps building	Bear validation Search & other decelerates toward low-single-digit or CPCs fall · FCF margin stays below ~18% as ~\$190B+ capex outruns cash · Cloud growth/margin rolls over · an adverse ad-tech (AdX/DFP) divestiture remedy	Reframe needed if generative-AI assistants take durable query share and ad load drops, the ~60%-of-revenue Search engine loses its economics - re-rate the terminal toward a slower-growth, lower-FCF-margin platform
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Forward-looking statements. Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.	Author may hold positions. The author may own, intend to acquire, or hold short exposure to \$GOOGL or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.	Do your own research. Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Search & other — Alphabet's core search-advertising revenue (~\$60.4B in Q1'26, ~60%+ of total); the swing-factor segment the generative-AI shift threatens.	Google Cloud — The cloud-infrastructure + Workspace segment (\$20.0B/+63% in Q1'26 at ~33% operating margin, ~\$462B backlog) - the second engine.	AI Overviews / AI Mode — Gemini-powered answers atop Search (2.5B / 1B MAU) monetizing roughly at Search-parity - how Google defends ad economics through the AI transition.
TPU — Tensor Processing Unit - Google's custom AI accelerator; serves Gemini and Cloud AI below merchant-silicon cost.	FCF margin — Free cash flow / revenue (~18% FY25, depressed by the capex ramp); whether ~\$190B capex converts to ~24% recovery.	